Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/26/2006
PI Name:	Kanas, Nick M.D.		
Project Title:	Crew Member and Crew-Ground Interactions During International Space Station Missions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Advanced human support technologies		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	nick.kanas@ucsf.edu	**Fax:**	FY 415-668-7503
PI Organization Type:	UNIVERSITY	**Phone:**	415-750-2072
Organization Name:	University of California-San Francisco/Northern California Institute for Research & Education		
PI Address 1:	Veterans Affairs Medical Center (116A)		
PI Address 2:	4150 Clement St.		
PI Web Page:			
City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2000 Space Biology 00-HEDS-01
Start Date:	07/31/1998	**End Date:**	02/28/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Increment 2-9 Data collection complete. Awaiting final report at the end of the cooperative agreement.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ritsher, Jennifer (University of California - San Francisco) Marmar, Charles (University of California - San Francisco) Weiss, Daniel (University of California - San Francisco)		
Grant/Contract No.:	NCC9-0161		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	During International Space Station (ISS) missions, international crews are engaged in complicated activities over long periods of time. A number of interpersonal issues likely to impact on these missions must be addressed in order to ensure healthy crewmember interactions and optimal performance. A review of the literature of space analog studies on Earth, anecdotal reports from previous space missions, and the principal investigator's own work involving astronauts and cosmonauts have isolated crew tension, cohesion, and leadership role as important interpersonal issues. Cultural and language differences among crewmembers and the relationship of space crews with monitoring personnel in mission control also are important factors that influence crew performance and well-being. Hypotheses related to these interpersonal issues were tested by having the crewmembers and personnel in ground control answer a series of questions from three standard mood and interpersonal group climate questionnaires, a critical incident log, and a culture and language questionnaire. The culture and language questionnaire was administered once before each mission. The other measures were completed on a weekly basis before, during, and after each mission and took approximately 15-20 minutes to fill out.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These findings will have direct application for the development of countermeasures to reduce displacement and decrements in mood and interpersonal interactions during the International Space Station missions and even longer missions, such as an expedition to Mars. In addition, the interpersonal interactions of long-duration, multi-national space crews constitute a laboratory of small group behavior that tells us a great deal about ways in which groups of people on Earth can relate with a minimum of tension and improved cohesion when they are under stress. Finally, the ability of people from previously opposing political blocks to engage in complex activities, such as undertaking a space mission or conducting research together, serves as a model for international cooperation on Earth.
Task Progress:	This year data collection was completed, data were compiled and analyses begun. Findings were reported in conference presentations and papers. Summary of results: Tension and negative emotions (dysphoria) in the crew environment was not found to change over time. Intra-crew tension and dysphora was found to displace onto outside monitoring personnel. Tension and dysphoria in the outside monitoring group was found to displace onto management. Crew cohesion was not found to change with the passage of time. Supportive leadership roles (but not task oriented leadership roles) of the mission commander were found to relate to crew cohesion. Changes in the leadership role of the mission commander were not found to occur over time. Russian-American differences were found: Russians reported more tension but less work pressure than Americans. Crew-Ground differences were found: ISS crewmembers reported having less negative emotions and more vigor and innovation then their colleagues in mission control. In general the ISS compared favorably to earth (US) based norms. ISS crewmembers reported less dysphoric emotions than normative earth samples and reported statistically higher levels of Cohesion and Task Orientation. Russian and American reports about the social climate on the ISS were similar with respect to norms and each other with the exception of Work Pressure. American crewmembers reported higher than norm-level Work Pressure. Russian crewmembers reported lower than norm-level Work Pressure.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Abstracts for Journals and Proceedings	Kanas N, Salnitskiy V, Ritsher J, Gushin V, Weiss D, Saylor S, Marmar C. "Crew and ground interactions during ISS missions: Conclusions and countermeasures. " 76th Annual Scientific Meeting of the Aerospace Medical Association: Charting the Course for the Future, Kansas City, MO, May 4-8, 2005. Aviat Space Environ Med. 2005 May;76(3):294. , May-2005
Abstracts for Journals and Proceedings	Kanas N, Salnitskiy V, Ritsher J, Gushin V, Weiss D, Saylor S, Marmar C. "Crew and ground interactions during ISS missions: Results. " 76th Annual Scientific Meeting of the Aerospace Medical Association, Charting the Course for the Future, Kansas City, MO, May 4-8, 2005. Aviat Space Environ Med. 2005 May;76(3):294. , May-2005
Abstracts for Journals and Proceedings	Kanas N, Salnitskiy V, Ritsher J, Gushin V, Weiss D, Saylor S, Marmar C. "Crew and ground interactions during ISS missions: Methods." 76th Annual Scientific Meeting of the Aerospace Medical Association: Charting the Course for the Future, Kansas City, MO, May 4-8, 2005. Aviat Space Environ Med. 2005 May;76(3):294. , May-2005
Abstracts for Journals and Proceedings	Kanas N, Salnitskiy V, Ritsher J, Gushin V, Weiss D, Saylor S, Marmar C. "Crew and ground interactions during ISS missions: Operational Issues." 76th Annual Scientific Meeting of the Aerospace Medical Association: Charting the Course for the Future, Kansas City, MO, May 4-8, 2005. Aviat Space Environ Med. 2005 May;76(3):293. , May-2005
Abstracts for Journals and Proceedings	Kanas N, Salnitskiy V, Ritsher J, Gushin V, Weiss D, Saylor S, Marmar C. "Crew and ground interactions during ISS missions: Background and Panel Overview." 76th Annual Scientific Meeting of the Aerospace Medical Association: Charting the Course for the Future, Kansas City, MO, May 4-8, 2005. Aviat Space Environ Med. 2005 May;76(3):293. , May-2005
Abstracts for Journals and Proceedings	Kanas NA, Salnitskiy VP, Ritsher JB, Gushin VI, Weiss DS, Saylor SA, Kozerenko OP, Marmar CR. "Psychosocial interactions during ISS missions." 15th IAA Humans in Space Symposium: Benefits of Human Presence in Space, Graz, Austria, May 22-26, 2005. 15th Humans in Space Symposium: Book of Abstracts, 2005, p. 21. , May-2005
Abstracts for Journals and Proceedings	Ritsher J, Kanas N, Ihle E. "Psychological adaptation and salutogenesis in space: Lessons from a series of studies." 15th IAA Humans in Space Symposium: Benefits of Human Presence in Space, Graz, Austria, May 22-26, 2005. 15th Humans in Space Symposium: Book of Abstracts, 2005, p. 22. , May-2005

Abstracts for Journals and Proceedings	Clement JL, Ritsher JB. "Cultural challenges faced by American mission control personnel working with international partners." 36th COSPAR Scientific Assembly, Beijing, China, July 16-23, 2006. 36th COSPAR Scientific Assembly, Beijing, China, July 16-23, 2006. Abstract Book. http://www.cosis.net/abstracts/COSPAR2006/02495/COSPAR2006-A-02495.pdf , Jul-2006
Abstracts for Journals and Proceedings	Ritsher JB, Kanas NA, Salnitskiy VP, Gushin VI, Weiss DS, Saylor SA, Marmar CR. "Cultural and crew-ground differences in mood and social climate: Results from two space station programs." 36th Scientific Assembly Committee on Space Research, Beijing, China, July 16-23, 2006. 36th Scientific Assembly Committee on Space Research, 2006. http://www.cosis.net/abstracts/COSPAR2006/02136/COSPAR2006-A-02136.pdf , Jul-2006
Abstracts for Journals and Proceedings	Kanas NA, Salnitskiy, VP, Ritsher JB, Gushin VI, Weiss DS, Saylor SA, Marmar CR. "Psychosocial issues on-orbit: results from two space station programs." 36th COSPAR Scientific Assembly, Beijing, China, July 16-23, 2006. 36th Scientific Assembly Committee on Space Research, 2006. http://www.cosis.net/abstracts/COSPAR2006/02130/COSPAR2006-A-02130.pdf , Jul-2006
Abstracts for Journals and Proceedings	Kanas N, Ritsher JB, Saylor SA. "Do psychosocial decrements occur during the 2nd half of space missions?" 57th International Astronautical Congress, Valencia, Spain, October 2-6, 2006. 57th International Astronautical Congress, 2006. , Oct-2006
Abstracts for Journals and Proceedings	Clement JL, Ritsher JB, Saylor SA, Kanas N. "Leadership challenges in ISS operations: Lessons learned from junior and senior mission control personnel." 57th International Astronautical Congress, Valencia, Spain, October 2-6, 2006. 57th International Astronautical Congress, 2006. , Oct-2006
Abstracts for Journals and Proceedings	Ritsher JB, Kanas N, Salnitskiy VP, Gushin VI, Saylor SA, Weiss DS, Marmar C. "Culture and language backgrounds of International Space Station personnel." 57th International Astronautical Congress, Valencia, Spain, October 2-6, 2006. 57th International Astronautical Congress, 2006. , Oct-2006
Abstracts for Journals and Proceedings	Robinson JA, Slack K, Baskin P, Trenchard M, Willis K, Ritsher JB. "Patterns in crew-initiated photography of Earth from ISS – is Earth observation a salutogenic experience?" 57th International Astronautical Congress, Valencia, Spain, October 2-6, 2006. 57th International Astronautical Congress, 2006. , Oct-2006
Abstracts for Journals and Proceedings	Clement JL, Ritsher JB, Saylor SA, Kanas N. "Leadership and cultural challenges in operating the International Space Station." Presented at the 77th Annual Scientific Meeting of the Aerospace Medical Association, Orlando, FL, May 14-18, 2006. Aviat Space Environ Med. 2006 Mar;77(3):276. , Mar-2006
Abstracts for Journals and Proceedings	Saylor SA, Ritsher JB, Kanas N, Weiss DS, Marmar CR. "Characterizing the social environment on board the International Space Station." Presented at the 77th Annual Scientific Meeting of the Aerospace Medical Association, Orlando, FL, May 14-18, 2006. Aviat Space Environ Med. 2006 Mar;77(3):276. , Mar-2006
Articles in Peer-reviewed Journals	Ritsher JB, Kanas N, Saylor S. "Maintaining privacy during psychosocial research on the International Space Station." Human Performance in Extreme Environments, 2006;8:21-6. , Oct-2006
Articles in Peer-reviewed Journals	Kanas N. "From Earth's orbit to the outer planets and beyond: Psychological issues in space." Acta Astronaut. 2011 Mar-Apr;68(5-6):576-81. http://dx.doi.org/10.1016/j.actaastro.2010.04.012 , Mar-2011
Articles in Peer-reviewed Journals	Kanas NA, Salnitskiy VP, Ritsher JB, Gushin VI, Weiss DS, Saylor SA, Kozerenko OP, Marmar CR. "Human interactions in space: ISS vs. Shuttle/Mir." Acta Astronautica, 2006 Jul-Sep;59(1-5):413-9. http://dx.doi.org/10.1016/j.actaastro.2006.02.007 , Jul-2006
Articles in Peer-reviewed Journals	Boyd JE, Kanas N, Gushin VI, Saylor S. "Cultural differences in patterns of mood states on board the International Space Station." Acta Astronautica, 2007 Oct;61(7-8):668-71. http://dx.doi.org/10.1016/j.actaastro.2006.12.002 , Oct-2006
Articles in Peer-reviewed Journals	Kanas NA, Salnitskiy VP, Ritsher JB, Gushin VI, Weiss DS, Saylor SA, Kozerenko OP, Marmar CR. "Psychosocial interactions during ISS misions." Acta Astronautica, 2007 Feb-Apr;60(4-7):329-35. http://dx.doi.org/10.1016/j.actaastro.2006.09.001 , Mar-2007
Articles in Peer-reviewed Journals	Ritsher JB, Kanas NA, Ihle EC, Saylor SA. "Psychological adaptation and salutogenesis in space: Lessons from a series of studies." Acta Astronautica, 2007 Feb-Apr;60(4-7):336-40. http://dx.doi.org/10.1016/j.actaastro.2006.09.002 , Mar-2007
Articles in Peer-reviewed Journals	Ihle EC, Ritsher JB, Kanas N. "Positive psychological outcomes of spaceflight: an empirical study." Aviat Space Environ Med. 2006 Feb;77(2):93-101. PMID: 16491575 , Feb-2006
Articles in Peer-reviewed Journals	Kanas N, Ritsher J. "Leadership issues with multicultural crews on the International Space Station: lessons learned from Shuttle/Mir." Acta Astronaut. 2005 May-Jun;56(9-12):932-6. PMID: 15835051 , May-2005
Articles in Peer-reviewed Journals	Ritsher JB, Ihle EC, Kanas N. "Positive psychological effects of space missions." Acta Astronaut. 2005 Jul-Oct;57(2-8):630-3. PMID: 16010762 , Jul-2005
Articles in Peer-reviewed Journals	Kanas N. "Interpersonal issues in space: Shuttle/Mir and beyond." Aviat Space Environ Med. 2005 Jun;76(6 Suppl):B126-34. PMID: 15943205 , Jun-2005
Articles in Peer-reviewed Journals	Ritsher JB. "Cultural factors and the International Space Station." Aviat Space Environ Med. 2005 Jun;76(6 Suppl):B135-44. PMID: 15943206 , Jun-2005

Papers from Meeting Proceedings	Kanas N, Ritsher J. "Psychosocial issues during a Mars mission." 1st Space Exploration Conference, Orlando, FL, January 30-February 1, 2005. AIAA Meeting Papers on Disc, CD Code: 1205-43717. Reston, VA : American Institute of Aeronautics and Astronautics, Inc., 2005. , Feb-2005
Papers from Meeting Proceedings	Kanas NA, Salnitskiy VP, Ritsher JB, Gushin VI, Weiss DS, Saylor SA, Kozerenko OP, Marmar CR. "Human interactions in space: ISS versus Shuttle/Mir." 56th International Astronautical Congress, Fukuoka, Japan, October 17-21, 2005. Paris : International Astronautical Federation, 2005. Paper #IAC-05-A1.5.02. , Oct-2005
Papers from Meeting Proceedings	Ritsher JB, Kanas N, Gushin VI, Saylor S. "Cultural differences in patterns of mood states on board the International Space Station." 56th International Astronautical Congress, Fukuoka, Japan, October 17-21, 2005. Paris : International Astronautical Federation, 2005. Paper #IAC-05-A1.5.03. , Oct-2005
Papers from Meeting Proceedings	Clement J, Ritsher JB. "Operating the ISS: Cultural and leadership challenges." 56th International Astronautical Congress, Fukuoka, Japan, October 17-21, 2005. Paris : International Astronautical Federation, 2005. Paper #IAC-05-A1.5.05. , Oct-2005